

Activity-3

Cruthika-B
192511232

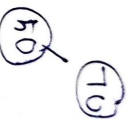
① Construct an AVL tree:

10, 50, 30, 90, 80, 130, 120

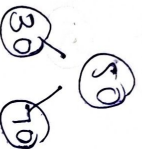
Insert 10



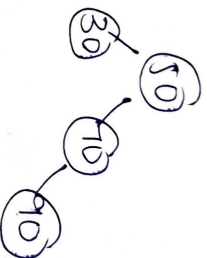
Insert 50



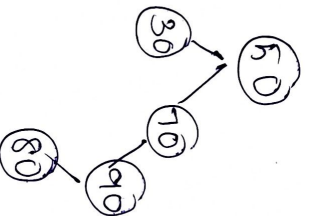
Insert 30



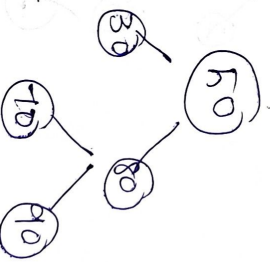
Insert 90



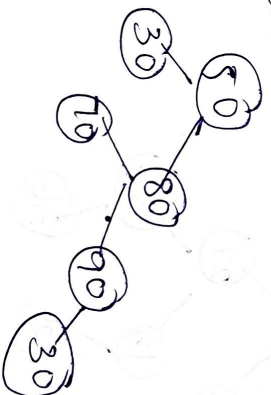
Insert 80



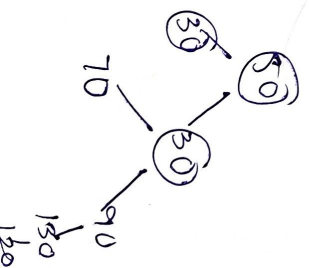
Rotation:



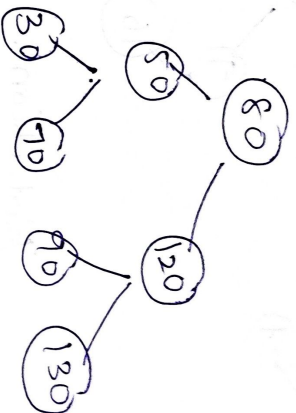
Insert 130



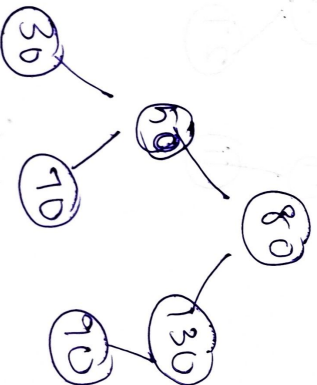
Insert 120



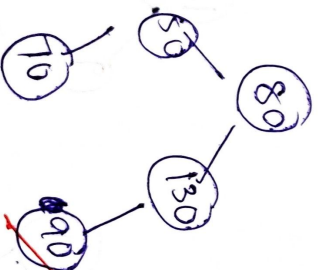
Rotation



delete 120



delete 30



2)

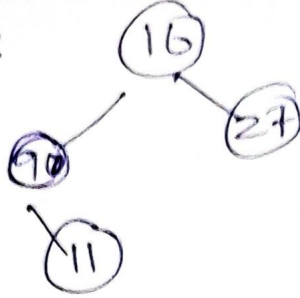
16, 27, 9, 11, 36, 54, 81, 63, 72

Step 3

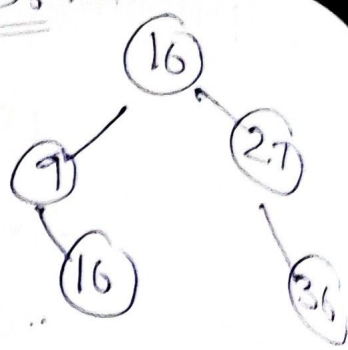
16

16

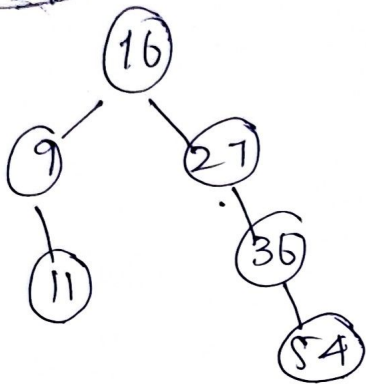
Step 4:



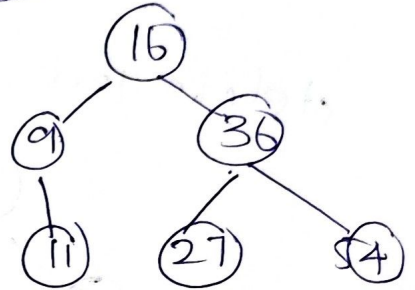
Step 5:



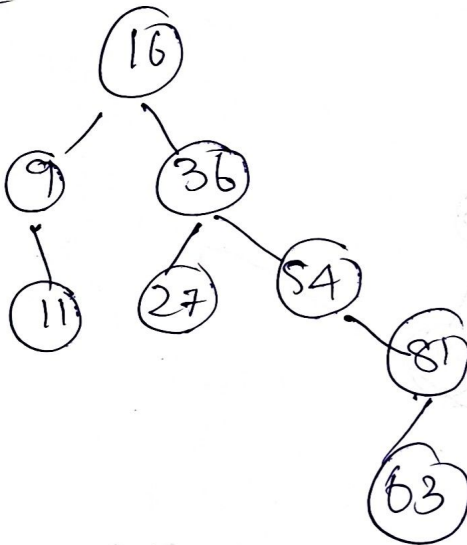
Step 6:



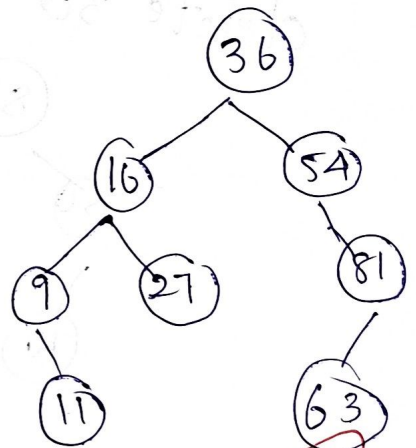
Rotation:



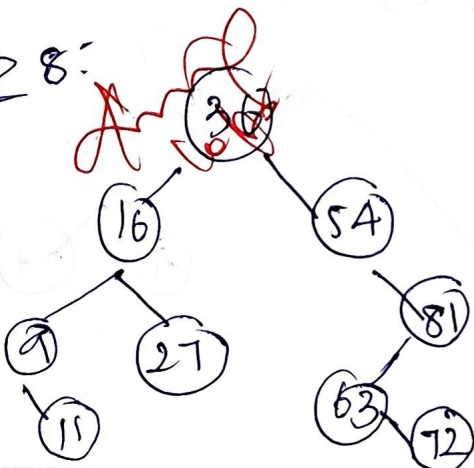
Step 7:



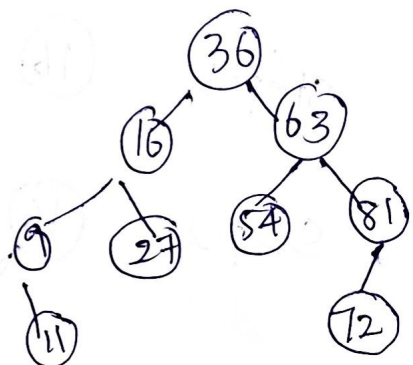
Rotation:



Step 8:



Final AVL tree:



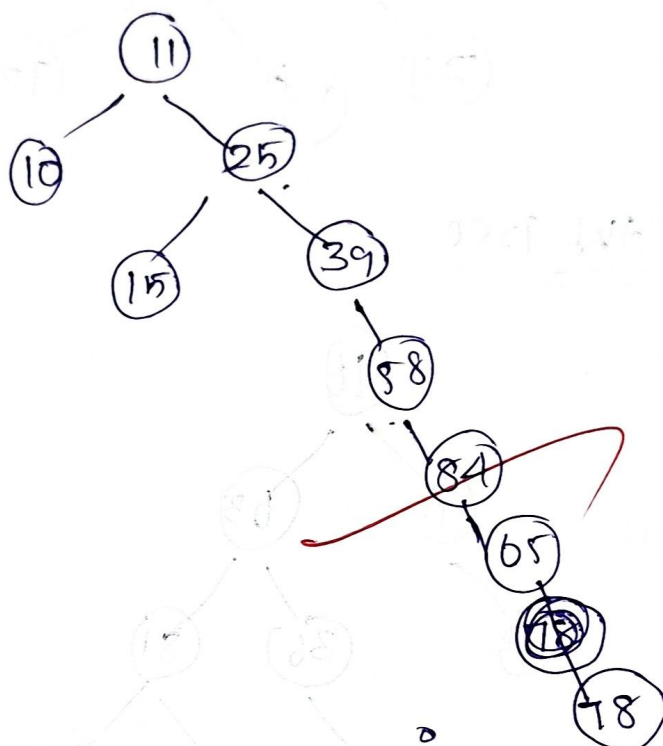
③ Create a Binary Search Tree

11, 25, 10, 15, 39, 58, 84, 65, 78

Smaller $\rightarrow$ left

Greater $\rightarrow$ right

Final Binary Search tree



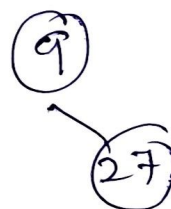
④ Binary tree after each insertion

9, 27, 50, 15, 2, 21, 36

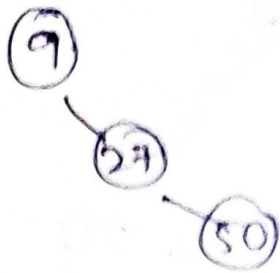
Insert 9



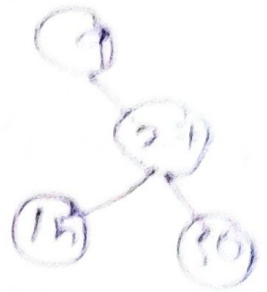
Insert 27



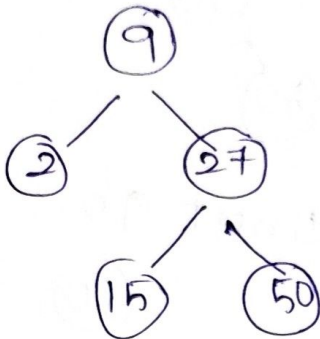
Insert 50



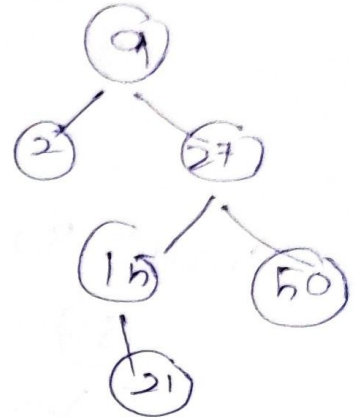
Insert 15



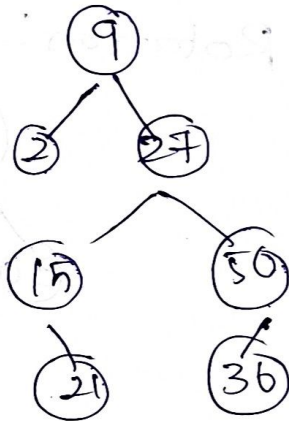
Insert 2



Insert 21



Insert 36



Final Answer:

